



1 The Challenge

1.1 The Scenario

- 1.1.1 There has been an accident at a manufacturing plant. There are a number of victims still trapped within the plant, and it is too hazardous to send in human rescue teams. Your autonomous robot must be able to navigate through a treacherous building which may have obstacles, uneven floors and restricted areas to identify victims and leave rescue kits to aid harmed victims. Time and technical skills are essential! Come and prepare to be the most successful Rescue Maze Response Team.

1.2 Event Divisions

- 1.2.1 The Rescue Maze Challenge has three divisions: Mighty Maisy Maze, Intermediate Maze and Open Maze.

Mighty Maisy Maze was developed with the aim to create a very accessible entry point for students who are new to the Challenge.

Intermediate Maze increases the difficulty of the Challenge to be a transition level between Mighty Maisy Maze and Open Maze. This division can also be used by Open Maze teams that are significantly changing hardware and/or software platform.

Open Maze extends the Challenge with more complexity for more experienced students.

Subject to age limitations and other criteria, Open Division teams who perform well in the National Event may qualify for the annual RoboCup Junior International Event. Other international events may have other divisions/qualification requirements.

The rules for each division are separated into their own sections with the chart below highlighting the differences:

Challenge Aspect	Mighty Maisy Maze	Intermediate Maze	Open Maze
Floor victims	X	X	
Wall victims			X
Rescue Kits for Harmed victims		X	X
Start/Finish using silver reflective tile	X	X	
Start/Finish using coloured tiles			X
Checkpoints at identified victims	X	X	
Checkpoints using silver reflective tiles			X
Maze: Allow use of A4 Reams for walls	X	X	
Maze: "No Go" black tiles	X	X	X
Maze: Floating Tiles		X	X
Maze: Debris		X	X
Maze: Speed Bumps		X	X
Maze: Obstacles			X
Maze: Stairs			X
Maze: Ramps			X
Maze: Multiple Levels			X
Maze: Tunnels			X
Exit Bonus	X	X	X
Count Bonus	X	X	



B Intermediate Maze

B1 Introduction

The robot needs to search through a maze for colour-identifiable victims. i.e. the robot need not find the fastest path through the maze, instead it should explore as much as possible of the maze. The robot will score points for each victim found. The robot should avoid areas with a black floor. If the robot is stuck in the maze, it can be ~~restarted at~~ **moved to** any previous correctly identified victim. If the robot can find its way back to the start tile after exploring the maze it will receive an exit bonus (see section B6.5). Additional count bonuses will be awarded upon exit if the robot can report the number and type of victims that it has identified in the maze (victims might be identified more than once) once at least half the victims have been identified (see section B6.6). If the robot can also deliver a Rescue Kit close to Harmed victims, it will earn additional points.

B2 Playing Field

All measurements in the rules have a tolerance of 5%.

B2.1 Maze Description

- B2.1.1 The maze will be on a single level and may consist of multiple distinct areas. Areas will have a horizontal floor and a perimeter wall. Walls can be constructed either using reams of A4 Paper laid on their long side or using the Open Maze walls.
- B2.1.2 Areas may be joined together by doorways.
- B2.1.3 Walls will be located approximately 29 mm (tolerance ± 15 mm) apart. There may be a 5mm gap between wall sections.
- B2.1.4 Doorways are approximately 290 mm (tolerance ± 15 mm) wide.
- B2.1.5 Walls that make up the maze are at least 150 mm to a maximum of 300 mm high. There may be a 5 mm gap between wall sections.
- B2.1.6 The walls **will be opaque and** can be any colour.

B2.2 Floor

- B2.2.1 Floors should be smooth (e.g. **melamine**) and may have up to 3 mm height difference at joints. There may be holes in the floor (about 5 mm diameter) for fastening walls. There may also be 5mm gaps between floor tiles. Floors will be a light colour and must not be a similar colour to silver reflective tiles, black tiles or victims.
- B2.2.2 The Start/Exit Tile will be marked by a silver reflective tile and enclosed with walls **on three sides**. The silver reflective tile will be fixed to the floor **but teams should be prepared for slight movements of these tiles**.
- B2.2.3 Throughout the Playing Field, there may exist black tiles that represent “no go” spaces. The black tile will be fixed to the floor **but teams should be prepared for slight movements of these tiles**.

B2.3 Path

- B2.3.1 ~~Walls will lead to the entrance/exit. Walls that lead to the entrance/exit are called linear walls. The walls that do NOT lead to the entrance/exit are called "Floating Walls".~~

Tiles that lead to the Start/Exit tile consistently following the leftmost or rightmost wall are called “Linear Tiles”. The tiles that do NOT lead to the starting tile consistently following the leftmost or rightmost wall are called ‘floating Tiles’.

Note: Black “no go” tiles will affect the determination of tile type since they can be considered virtual walls



B2.3.2 Paths will be approximately 290 mm (tolerance ± 15 mm) wide but may open into foyers wider than the path.

B2.4 Debris, Speed Bumps, Obstacles and Stairs

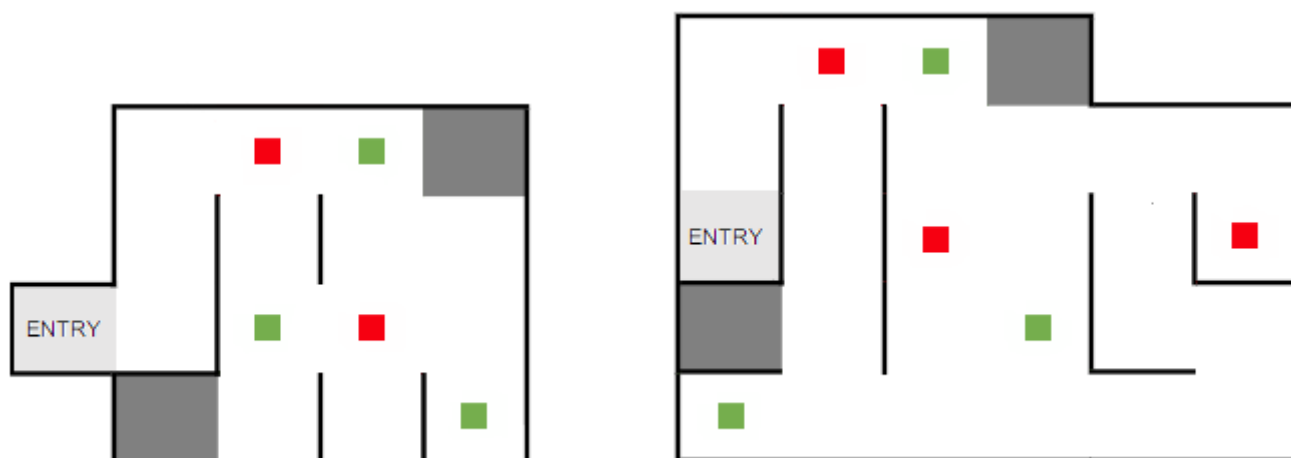
B2.4.1 **Speed Bumps** are fixed to the floor, have a maximum height of 5 mm and will be no closer than 50 mm apart.

B2.4.2 **Debris** will not be fixed on the floor and will have a maximum height of 5 mm. Debris can be placed anywhere on the tile and must be a contrasting colour to the victims, the start tiles, black tiles.

Hint: Make sure that your robot can negotiate speed bumps and has sufficient ground clearance.

B2.4.3 There are no Obstacles, Stairs or Ramps in the Intermediate Maze division.

Example Courses:



B2.5 Environmental Conditions

B2.5.1 This event simulates a disaster area. There may be gaps and misalignments between wall panels, differences in flooring materials and heights. Teams should expect the environmental conditions at a tournament to be different from the conditions at their home practice field.

B2.5.2 Teams must come prepared to adjust their robots to the conditions at the venue.

B2.5.3 Lighting and magnetic conditions may vary within and between Playing Fields at the venue.

~~B2.5.4 The arena may be affected by magnetic fields (e.g. generated by under floor wiring and metallic objects).~~

~~B2.5.5 Teams should prepare their robots to handle unexpected lighting interference. While the organizers and referees will try their best to minimize external lighting interference, it is not possible for them to foresee all unexpected interferences such as camera flash from spectators.~~

~~B2.5.6 The Organizing Committee will try their best to fasten the walls onto the field floor so that the impact from regular robot's contact should not affect the robot. All measurements in the rules have a tolerance of 5%.~~

~~B2.5.7 Objects to be detected by the robot will be distinguishable from the environment by their colour.~~

B3 Victims

B3.1 Definition

B3.1.1 Victims are represented by red or green coloured 50 mm squares fixed to the floor ~~approximately 150 mm from the nearest wall (as measured to the centre of the square).~~



B3.1.2 Unharmred victims are depicted as green squares while Harmed victims are red squares.

B3.2 Locations

B3.2.1 There will be a minimum of four (4) victims in any round.

B3.2.2 Victims will be fixed to the centre of a tile.

B3.2.3 Victims can be located anywhere within the maze except on black tiles and Start/End tiles.

~~B3.2.4 Victims will be placed more than 100 mm apart as measured from the edge of each square.~~

B.4 Rescue Kits

B4.1 A Rescue Kit represents a basic health package distributed to a harmed victim caught in a natural disaster. It symbolises tools or devices used in the rescue process, such as GPS Transponders or even something as simple as food, water or light source providers.

B4.2 Each Rescue Kit must have a minimum volume of 500 mm³ (e.g. 8 mm x 8 mm x 8 mm).

B4.3 Each team can only carry a maximum number of 12 of those kits.

B4.4 Each team is responsible for the whole Rescue Kit system (the maximum of 12 kits), including bringing the rescue kits to the event. The Robot Handler is responsible for loading their own Rescue Kits on their robots and cleaning the field with the referee's/judges' authorization after the game is called to end.

B.5 Game Play

B5.1 Pre-round Practice

B5.1.1 Where possible, competitors will have access to practice arenas for calibration, testing and tuning throughout the event.

Note: The lighting and magnetic conditions in the area of the practice mazes may vary from the competition mazes.

B5.1.2 Whenever there are dedicated independent arenas for event and practice, it is at the organizers' discretion if testing is allowed on the event arena.

B5.2 Humans

B5.2.1 Teams should designate one of its own team members as Robot Handler. Only this team member will be allowed access to the practice/event arenas, unless otherwise directed by a referee. Only the robot handler will be allowed to interact with the robot during a scoring run.

B5.2.2 The robot handler can move the robot only when told to do so by the referee.

B5.2.3 Other team members (and any spectators) within the vicinity of the rescue arena have to stand at least 1500 mm away from the arena while their robot is active, unless otherwise directed by the referee. **Non-team members are not allowed to communicate with the team during a round.**

B5.2.4 No one is allowed to touch the Playing Fields intentionally during a scoring run.

B5.3 Start of Play

B5.3.1 A run begins at the scheduled starting time whether or not the team is present/ready.

B5.3.2 Once the scoring run has begun, the playing robot is not permitted to be taken from the event area for any reason.



- B5.3.3 Each run lasts a maximum of 240 seconds and includes time for calibration. Time Limits for a round may be varied at the discretion of the Challenge Coordinator at the event.
- B5.3.4 Calibration is defined as the taking of sensor readings and if necessary, modifying a robot's program to accommodate such sensor readings. Once the clock has started, a team may calibrate their robot at as many locations as desired on the arena, but the clock will continue to count down.
- B5.3.5 Calibration time is not for pre-mapping the arena and/or the locations of the victims. Pre-mapping activities will result in immediate robot disqualification for the round.
- B5.3.6 The maze will have one silver reflective start tile.

B5.4 Scoring Run

- B5.4.1 Modifying a robot during a run is prohibited, which includes remounting parts that have fallen off, unless a Restart is declared.
- B5.4.2 All parts that the robot loses intentionally or unintentionally will be left in the arena until the run is over, unless a Restart is declared. Neither the team nor the judge are allowed to remove parts from the arena during a run.
- B5.4.3 A "visited tile" means that more than half of the robot is inside the tile when looking down from above and shall be determined by the referee.
- B5.4.4 The scoring run ends when:
- B5.4.4.1 The time expires.
 - B5.4.4.2 The Robot Handler declares an end of their scoring run. The team will be awarded all points achieved up to the call for end of round.
 - B5.4.4.3 The robot returns to the start tile and gets the exit bonus. To collect points, the robot must stop on the start/exit tile **such that the entire robot is contained on the tile** and indicate in some way that it has completed the round. Exit indication should be distinct from the victim indication as detailed in section B6.2. The operator should be prepared to demonstrate the exit indication to the referee if required.
 - B5.4.4.4 A team member touches the arena or their robot without permission from a referee. If a team member is in breach of this rule the referee can declare a Restart where the robot is returned to the Start tile as per B5.6.1.

B5.5 Lack of Progress

- B5.5.1 A Lack of Progress occurs when:
- B5.5.1.1 The Robot Handler requests a Lack of Progress from the referee provided that the robot can genuinely not continue through either a ~~stoppage or recursive (looping) behaviour~~ **stoppage of three (3) or more seconds, or behaviour of three (3) or more loops.**
 - B5.5.1.2 A robot fails to retreat from a "visited" black tile. A robot is deemed to have entered the tile when more than half the robot is within the tile as determined by the referee. For a successful retreat, it needs to back up without turning inside the black tile (it has to move straight backwards inside of a black tile). (See definition of visited tile on rule B5.4.3.) If a robot is deemed to have visited the black tile, a Lack of Progress will be called as per rule B5.5.2.
 - B5.5.1.3 A robot or a team member damages the arena.
- B5.5.2 If a Lack of Progress occurs the robot must be returned to any previously visited checkpoint (or the Start tile if the robot has not reached a checkpoint). The robot can be placed in any direction but must be wholly within the tile. The program can be paused ~~but not reset or changed~~ **or reset but not changed**. All points scored to that stage are still valid, but debris will not be reset.



Note: teams should consider the implications of resetting their code on stored variables such as victim counts

B5.6 Restart

B5.6.1 After a Lack of Progress or when Rule B5.4.4.4 is breached, the Robot Handler may declare a **Restart**. After a Restart, the team can reset the power supply (turn the robot off and on), change programs, remount parts that have fallen off, and the maze will be returned to original condition by the referee.

Note: All points earned prior to a call of restart are invalid and will be reset.

Note: The round timer remains running.

B.6 Scoring

B6.1 **Successful Victim Identification:** Robots are rewarded points for each Successful Victim Identification in the arena:

- 10 points per Unharmd victim.
- 25 points per Harmed victim, with an additional 10 points for a successful rescue kit deployment.

B6.2 A robot can carry out the following action to successfully identify a victim by stopping for at least 1 second over the victim and clearly indicating by either visual indicator or sound.

Hint: Sounds can be hard to hear in a loud venue, so a visual indicator is recommended.

B6.3 **No duplicate rewards.** Each victim can only be identified and scored once. Points for a successful rescue kit deployment can only be scored once.

B6.4 Once correctly identified a victim can be used as a checkpoint for lack of progress.

B6.5 **Exit Bonus.** A successful exit bonus of 25 points will be awarded if the robot reaches and stops on the Start/Exit Tile **such that the entire robot is contained on the tile**.

B6.6 **Count Bonus.** Once at least half of the total victims in the maze (rounding down) have been identified, the robot will be awarded a further bonus of 25 points if it stops on the Start/Exit tile and clearly indicates the number of Unharmd victims the robot has found. A bonus of 25 points will be awarded if the number of Harmed victims the robot has found is clearly indicated. There is no penalty if victims are identified more than once.

Note: the victim should not be detected more than once per time that the sensor passes over it.

For example, if there are 2 Harmed victims, but the robot has successfully identified 3 (i.e. it has counted one victim twice) then the Harmed victim count should be 3. Teams are advised to consider how their counted variables may be affected if restarting the program in the event of a lack of progress.

B6.7 **Scoring Summary:**

State	Indicator	Points
Unharmd victim found	Stop for at least 1 second and give clear indication of victim identification	10 points per victim
Harmed victim found	Stop for at least 1 second and give clear indication of victim identification	25 points per victim
Exit Bonus	The robot will be awarded 25 points for reaching and stopping on the Start/Exit tile such that the entire robot is contained on the tile within Time Limit	25 points
Unharmd Victim Count Bonus	The robot will be awarded 25 points for a count of Unharmd victims the robot found	25 points



Harmed Victim Count Bonus	The robot will be awarded 25 points for a count of Harmed victims the robot found	25 points
Rescue Kit deployed for Harmed victims	Drop one rescue kit on the tile of an identified Harmed victim	10 points per victim

B6.8 **Successful rescue kit deployment.** Robot should drop a rescue kit on the tile where an identified Harmed victim is. The deployment point and the rescue kit need to be wholly on the tile of the victim. The robot is awarded points (see table) per successful rescue kit deployment.

B.7 Event Structure

- B7.1 The regular event will consist of as many preliminary rounds as possible with the team scores cumulative over the rounds. Placings at the end of the preliminary rounds will be awarded on the basis of the cumulative scores. In the event of a tied position the placings will be awarded based on the number of victims identified over the preliminary rounds. If still tied, the placings will be decided based on the lowest cumulative time over the preliminary rounds.
- B7.2 The organisers may include Super Rounds for the final rounds of the event.
- B7.3 The top 4 scoring teams in the preliminary rounds will be eligible to compete in the Super Rounds.
- B7.4 Placings at the end of the event will be awarded based solely on the cumulative scores of the Super Rounds. In the event of a tied position the placings will be awarded based on the number of victims identified over the Super Rounds. If still tied, the placings will be decided based on the lowest cumulative time over the Super Rounds. If still tied, the placings will then refer to the preliminary rounds results.
- B7.5 In the Super Rounds all teams compete at the same time, on separate independent courses.
- B7.6 The teams will select their initial ~~starting tile~~ course in the order that they have finished in the preliminary rounds.
- B7.7 After each Super Round, the competing teams will start the next round from the next ~~start tile~~ course moving clockwise.
- B7.8 Courses for the Super Rounds must be independent but can have missing walls as long as there is a black tile to prevent robots from crossing into another course.



2 Robot

2.1 Robot Configuration

There are some robot configuration rules that are consistent across all Challenges. Please ensure you read the General Rules, which can be downloaded from the [Rescue Maze Challenge Page](#) on the RoboCup Junior Australia Website, for robot configuration requirements.

2.1.1 The height of a robot must not exceed 300 mm.

Care should be taken in robot design to ensure that it is of a manageable size to negotiate its way successfully through the maze and past obstacles.

2.1.2 Robots may not have any sensor or devices that enable it to 'see' over the walls.

2.1.3 Any robot kit or building blocks, either available on the market or built from raw hardware and materials, may be used, as long as the design and construction are primarily and substantially the original work of the students.

2.1.4 Any commercially produced robot kits or sensors components that are specifically marketed to complete any single major task of RoboCup Junior Australia Rescue Maze will be disqualified. If there is any doubt, teams should consult the Technical Committee (TC).

2.1.5 Robots should be well engineered and constructed. The robot should not fall apart during the game. If the robot has substantially failed mechanically, fallen apart or is unable to complete the challenge, the robot will be deemed damaged, and the Robot Handler will be asked to remove the robot from the field. The Robot Handler may choose to restart according to sections A5.6.1, B5.6.1, C5.6.1, otherwise the round is deemed to have ended.

2.1.6 Each team is permitted to compete with one (1) robot only for the duration of the competition. The same robot must be used for all runs and rounds of the event.

Teams may make repairs, adjustments, and incremental improvements to their robot during the competition, provided the robot continues to comply with all competition rules and requirements.

Replacing the robot with a different robot during the competition is not permitted. Significant modifications or major rebuilds that substantially alter the robot's structure, drive system, or overall design must be approved by the judges or Organising Committee before the robot is permitted to compete again.

2.1.7 A team will not be able to compete with a robot substantially the same as another team's robot.

2.1.8 A team will not be able to compete with a robot that is identical to another team's robot from previous years.

~~2.1.9 For the safety of participants and spectators, only lasers of class 1 and 2 are allowed. This will be checked during inspection.~~

2.2 Robot Control

2.2.1 Robots cannot be started from a secondary device, such as a laptop, tablet, or mobile phone. Robots must have their program downloaded to them and be able to be started/restarted manually by the Robot Handler.

2.2.2 Robots must be autonomous in operation. If the robot has the capability for remote or any other wireless control (such as by Bluetooth, Wi-Fi or another form of wireless communication), the team must prove that they have disabled the capability for third party operation in some way. This could be by software, hardware or degree of human interaction. Robots that do not comply may face immediate disqualification from the event. Distributed control is allowed but must operate without human interaction after the robot has started the round.

2.2.3 Pre-mapping and/or any type of dead reckoning (i.e., configuring the robot's program based on predefined locations, tiles, obstacles, location of victim, number of victims etc. before game play) is prohibited.



2.2.4 The use of remote control of any kind is forbidden.

2.3 Inspection

2.3.1 Teams may be scrutinised to establish the ownership of robot design and programming.

2.3.2 The robot will be inspected by a panel of referees before/during or after the event to ensure that the robot adheres to all relevant rules.

2.3.3 It is the responsibility of teams to have their robot re-inspected if their robot is modified at any time during the event.

2.4 Violations

2.4.1 Any violations of the inspection rules will prevent the robot from competing in a round until modifications have been made to the robot to ensure compliance.

2.4.2 Modifications must be made within the time schedule of the events. Rounds will not be delayed due to late teams.

2.4.3 If a robot fails to meet all specifications (including modifications) the robot will be disqualified from that round (but not the event).

2.4.4 If it is determined that the work on the robot is not substantially the original work of the team members or the construction or programming of the robot be deemed not that of the team, the team will be referred to officials. Should the Rescue Coordinator (or other relevant person) on the advice of the officials then uphold the view of the scrutiniser, the team may be disqualified from the event.



3 Validation of Work

3.1 Electronic Submission

Each team is expected to submit two (2) digital elements prior to competition based on dates provided by the organisers:

- a. Annotated Code
- b. Digital Poster

3.1.1 Annotated Code Submission

3.1.1.1 Teams must submit a copy of the most recent version of their competition code. The code must be clearly annotated with comments that explain the purpose and function of key sections of the program, including (where applicable) sensor use, decision-making logic, and actuator control.

Note: Teams are permitted to continue updating their code after the digital submission deadline, but must maintain an up-to-date version of their annotated code. Students may be asked to show this updated code to judges upon request.

3.1.1.2 The submitted code must accurately reflect the robot used in competition.

3.1.1.3 Teams who fail to submit their annotated code will not be permitted to compete.

NOTE: This can be submitted as a power point, screen shots with annotations, doc, etc..

3.2.2 Digital Technical Poster Submission

3.2.2.1 Each team must submit one (1) poster that communicates their team and robot design if the team wishes to be considered for special awards.

Note: this poster may be shared publicly. Teams must not include any information that can't be publicly shared (e.g. student personal information other than first name, photos that would allow personal identification, etc.).

3.2.2.2 The poster must be A3 size and must include all of the following information:

- 1 Team name
- 2 Team member first names and roles
- 3 One interesting/innovative feature of the robot's design or engineering
- 4 A challenge encountered during development and how it was overcome
- 5 What the team has learned through participation in the competition

3.2.2.3 The poster must be clear, well-presented, and suitable for display. Teams may generate the poster using any software they chose (PowerPoint, Canva, Paint, etc.) or make a physical poster and submit a photo/scan of the poster. The poster must be the team's own original work.

3.2.2.4 Teams who fail to submit a poster will not be eligible for special awards.

Note: Teams are encouraged to bring a physical copy of their poster to display at the competition, however this is not mandatory.

3.3 Interviews

3.3.1 Teams may be required to attend a technical interview to explain the operation of their robot to verify that the design, construction and programming of the robot is the students' work. There are no set questions. If



interviews are being conducted, either a schedule will be released, or teams will be advised to go for an interview throughout the event well in advance of the Head-to-Head Round.

- 3.3.2 The format, timing, and selection of teams for interview will be determined by the Organising Committee. Teams will be required to bring an up-to-date version of their annotated code to any requested interviews.
- 3.3.3 Interviews are not scored and do not contribute to overall team score.
- 3.3.4 Team member(s) will be asked questions about their preparation efforts, and they may be requested to answer surveys and participate in videotaped interviews for research purposes.



4 Teams

There are some Team rules that are consistent across all Challenges. Please ensure you read the General Rules, which can be downloaded from the [Rescue Maze Challenge Page](#) on the RoboCup Junior Australia Website, for division rules that are consistent across all Challenges.

4.1 Definition

- 4.1.1 The eligibility requirements for each Rescue Maze Challenge Division are (Students should be guided to a higher division by their mentor as capability increases):
 - 4.1.1.1 **Mighty Maisy Maze:** This is open to students who are new to Rescue. Teams will only be allowed to enter this division where all students have less than two years' experience participating in Rescue Maze (if all Team Members are in Year 8 or below), or have not participated in Rescue Maze in a prior year (if any Team Member is in Year 9 or above).
 - 4.1.1.2 **Intermediate Maze:** Open to students at an age that would typically be studying at a recognised primary or secondary provider. This is open to all students with less than two years' experience in Intermediate Maze or students attempting a new challenge such as creating a robot based on an Arduino or Raspberry Pi controller programmed with a text-based coding language.
 - 4.1.1.3 **Open Maze:** Open to all students at an age that would typically be studying at a recognised primary or secondary provider.
- 4.1.3 For each round, one team member is to be nominated as the Robot Handler. Only the Robot Handler is permitted to enter the Game Zone and handle the robot during the round. All other team members must remain outside the Game Zone unless authorised by the Referee, Official or other relevant person.
- 4.1.4 The Robot Handler is the only team member permitted to communicate directly with the Referees, and Officials.